FILIPPO MARCANTONI

Robotics Engineer | Computer Vision, Robot Learning & Physical AI

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SUMMARY

M.S. Robotics Engineer with experience building end-to-end systems across 3D computer vision, robot learning, generative AI, swarm robotics, and industrial automation. Focused on bridging perception, learning, and control for real-world deployment.

EDUCATION

Worcester Polytechnic Institute (WPI) — *Worcester, MA, USA*

M.S. Robotics Engineering | GPA 3.90 — *Aug 2024 – May 2026*

B.S. Robotics Engineering | GPA 3.89, Summa Cum Laude — *Aug 2021 – May 2025*

Coursework: ML for Robotics, Computer Vision, Robot Dynamics & Control, Swarm Intelligence, HRI, Embedded Systems.

EXPERIENCE

AI & Industrial Robotics R&D Intern | B&R Industrial Automation, Austria — *Jun 2025 – Aug 2025*

- Built an LLM-based code-generation copilot for B&R Automation Studio using Transformers, RAG, and Azure AI retrieval, producing Motion ST templates for industrial robot programming.
- Redesigned the retrieval backend with a domain-specific schema and a syntax-aware chunking pipeline for industrial robotics ST code, enabling more relevant context retrieval across large code corpora.

Machine Vision & AI R&D Intern | Makro Labelling SRL, Italy — *May 2024 – Aug 2024*

- Developed C++ and Python machine-vision pipelines for bottle orientation, label alignment, and defect inspection on high-speed labeling lines, combining segmentation, feature matching, and homography estimation.
- Benchmarked classical and deep-learning models across bottle formats, lighting, and label shifts to characterize failure modes for automated quality control.

RESEARCH

STL-to-Fluoroscopy 2D–3D Registration | FuTURE Lab, WPI — *Aug 2025 - Dec 2025*

- Designed a self-supervised fluoroscopy-to-volume registration workflow using DiffDRR, 3D Slicer, and a DICOM pipeline for image-guided endovascular procedures; validated patient-specific pose estimation on a 3D-printed aorta phantom without ground-truth labels.

PiELo: Reactive Swarm Programming Language (MQP) | NEST Lab, WPI — *Aug 2024 – May 2025*

- Designed a reactive swarm-programming language with consensus and virtual stigmergy as first-class primitives; validated in ARGoS3 and on Khepera IV with Vicon tracking; co-authored peer-reviewed publication.

VR-Based Multimodal Teleoperation Interface (Capstone) | HiRo Lab, WPI — *Jan 2025 - May 2025*

- Built a Unity + ROS teleoperation system for a robotic nursing assistant with Meta Quest head-motion mirroring, real-time multi-camera streaming, Whisper voice commands, and ZED-Mini object-detection-driven camera selection.

SELECTED GRADUATE PROJECTS

Robot Learning – RL, A3C, Behavioral Cloning & Diffusion Policy *(RBE 577 – Machine Learning for Robotics, Spring 2026)*

- Implemented policy iteration, REINFORCE, A2C, and A3C for LunarLander and PyBullet Kuka vision-based grasping, reaching 55% grasp success with a CNN actor-critic policy.
- Trained robomimic policies on 59 PickPlaceCan demonstrations: LSTM Behavioral Cloning raised success from 0.02 to 0.80; Diffusion Policy with low-dimensional and RGB observations further improved success to 0.90.

Classical MSCKF & Deep Visual-Inertial Odometry *(RBE/CS 549 – Computer Vision, Spring 2026)*

- Implemented a stereo MSCKF VIO backend with IMU propagation, camera-state cloning, stereo feature tracking, and nullspace EKF updates; evaluated on EuRoC and RPG datasets.
- Trained IMU-only, vision-only, and fused deep VIO networks on synthetic Blender trajectories using geodesic SO(3) and trajectory-composition losses.

Einstein Vision – Monocular Autonomous-Driving 3D Visualization *(RBE/CS 549 – Computer Vision, Spring 2026)*

- Built a Tesla-inspired monocular perception pipeline integrating Detic, YOLOv8/DETR, Mask R-CNN, ZoeDepth/MiDaS, YOLO-3D, PyMAF, and RAFT.
- Reconstructed 3D driving scenes in Blender with lanes, vehicles, pedestrians, traffic objects, orientation, motion state, and collision-risk estimation.

3D Scene Reconstruction – Classical SfM & NeRF *(RBE/CS 549 – Computer Vision, Spring 2026)*

- Implemented Structure-from-Motion from scratch: SIFT tracking, RANSAC, 8-point fundamental matrix, essential-matrix recovery, cheirality, PnP-RANSAC, triangulation, and bundle adjustment.
- Built a PyTorch NeRF with positional encoding, 5D MLPs, hierarchical sampling, and differentiable volume rendering.

MaxCal-Derived Swarm Control *(RBE 511 – Swarm Intelligence, Spring 2026)*

- Developed a Maximum-Caliber decentralized swarm controller deriving local Markov transition rules from coverage and information-diffusion constraints, integrating inverse MaxCal solving, and quadratic mode selection for state-driven oscillation.

TECHNICAL SKILLS

Programming: Python, C++, C, C#, MATLAB, Java, JavaScript, Buzz, DrRacket.
Robotics & Simulation: ROS/ROS2, Gazebo, MuJoCo, PyBullet, ARGoS3, robosuite, robomimic, Unity, Blender, Vicon, VR/XR, Dynamixel, ZED SDK, Automation Studio, 3D Slicer.
Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, CUDA, Transformers, LLMs, RAG, Reinforcement Learning, Imitation Learning, Behavioral Cloning, Diffusion Policy, physics-informed learning.
Computer Vision & 3D Perception: OpenCV, SfM, NeRF, Visual-Inertial Odometry, SLAM, MSCKF, RANSAC, PnP, bundle adjustment, homography, segmentation, object detection, monocular depth, optical flow, point cloud, volumetric rendering.
Tools & Infrastructure: Git, Docker, Linux, Jupyter, Simulink, SolidWorks, Fusion 360, Azure AI, AWS, PostgreSQL, MySQL, HPC/Slurm, CI/CD.
Languages: Italian (native), English (fluent), Spanish (advanced).